

UCC611: BUILDING INFORMATION MODELLING IN CONSTRUCTION

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Course Objectives:

The objective of this course is to provide students with a comprehensive understanding of Building Information Modeling (BIM) in civil and structural engineering. The course covers BIM standards, data integration, and modeling workflows to enhance collaboration across disciplines. Students will gain expertise in modeling concrete, steel, and precast structures, along with advanced scheduling and detailing techniques. The curriculum also introduces cloud-based worksharing, BIM model coordination, and clash detection to ensure efficient project execution. Additionally, the integration of 4D (time) and 5D (cost) BIM, along with infrastructure design using industry-relevant software.

BIM STANDARDS AND PROCESSES FOR CIVIL AND STRUCTURAL ENGINEERING

Base data for civil and structural BIM, Infrastructure Structure modelling workflows; Federation of structural models with models from other disciplines and the legal implications; Family creation / Intelligent parametric libraries; Data/information exchange with structural analysis software.

CONCRETE, STEEL AND PRECAST MEMBERS MODELING

Advanced modelling technique, scheduling of precast concrete elements, rebar detailing, and creation of schedule.

PHYSICAL & CLOUD WORKSHARING

Introduction of physical and cloud worksharing methodologies within a common data environment. Integrate and real-time collaborate with Architecture Engineering and Construction.

BIM MODEL COORDINATION AND CLASH DETECTION

Modelling of RCC Structures and Prefabricated components for clash detection. Identification and resolution of clashes or conflicts in a 3D digital model, ensuring seamless collaboration among various disciplines involved in Civil Engineering project

4D, 5D BIM

Integration of time (4D) and cost (5D) dimensions within the BIM process in the project.

DESIGN AND MODELING OF ROADS

Build models automatically from maps, road design and traffic simulation, Simulation of drainage systems, bridges, watersheds, and slopes. Conceptual and detailed design using BIM based software and InfraWorks together. Design intersections, road widenings, and diverse road cross-sections. Conceptually and precisely design roads and buildings.

Project: Hands-on experience in modelling, articulating, perform tasks and collaborate in a project.

Course learning Outcome (CLO):

Upon completion of this course, the students will be able to:

1. Apply advanced modeling techniques for reinforced concrete, steel, and precast elements, including scheduling, rebar detailing, and element management.
2. Integrate physical and cloud-based work-sharing methodologies within a common data environment to enhance collaboration in AEC projects.
3. Utilize BIM tools for project time (4D) and cost (5D) in construction projects.
4. Apply BIM methodologies for road design, traffic simulation, drainage systems, and landform modeling using tools.

Reference Material

1. H. Savla, C. Chavan, P. Patil., “*Building Information Modeling: Global & Indian Perspective*”. Notion Press Publisher, 2021
2. K. Bava& B. Sinha Building Information Modeling, Bio-Green Books, 2023

EvaluationScheme:

Sr. No.	EvaluationElements	Weightage (%)
1	MST	25
2	EST	35
3	Sessional(Tutorials & Project)	40